

Alexandros Nikolaou

+49-17636962587 | alexandros.nic@hotmail.com

Portfolio | LinkedIn | Berlin, Germany

PROFESSIONAL SUMMARY

Robotics Systems Engineer specialized in bridging the **Sim2Real** gap for autonomous robots and industrial manipulators. Experienced in **low-level middleware optimization** (Linux Kernel, RDMA), **embodied AI** and **simulations**. Proven track record of accelerating market entry for startups, leading international deployments in complex, unstructured environments and optimizing real-time safety-critical systems.

TECHNICAL SKILLS

Robotics & AI: ROS 1 & 2, SLAM, Navigation Stack, MoveIt2, Omniverse, Isaac Lab, Isaac Sim, MuJoCo, Gazebo, Webots, VLA, VLM, LLM, AI Agents, PyTorch, Keras, Tensorflow, OpenCV, Reinforcement Learning
Systems & Software: C++, Python, CUDA, Linux Kernel, RDMA, Docker, Wireshark, VTune, perf, Git
Engineering & Design: Control Systems, SolidWorks, ANSYS (FEA), 3D Printing (FDM & Powder-based), Simulink, Matlab
Languages: Greek (Native), English (C1), German (A1.2)

EXPERIENCE

Freelancer

Berlin, Germany

Robotics & AI Engineer

Oct 2025 – Present

- Architecting and deploying custom AI agent frameworks to automate complex workflows for industry clients.
- Developing digital twins and robotic simulations in NVIDIA Isaac Sim.

Conbotics

Berlin, Germany & Singapore

Robotics Deployment Engineer

Apr 2025 – Present

- Successfully executed on-site robot deployments in Singapore; directly instrumental in converting the client into Conbotics' lead investor, securing over €1M in funding.
- Managed high-stakes demonstrations for executive leadership, including the CEO of a multi-billion dollar enterprise and the Minister of Innovation, while resolving critical hardware and software discrepancies daily.
- Engineered a containerized deployment framework to automate simulation spawning for CI/CD unit testing, significantly streamlining the Sim2Real validation process.

Bosch

Magdeburg, Germany

Software Engineer - Autonomous Driving

Mar 2023 – Oct 2024

- Boosted sensory data throughput by 10% and reduced latency in the autonomous vehicle stack using C++, CUDA, and shared memory optimization.
- Engineered for extreme performance at the kernel level by conducting microsecond-level bottleneck analysis and network packet debugging within the Linux kernel and RDMA subsystems.

University of Potsdam

Potsdam, Germany

Robotics Research Engineer

Feb 2022 – Nov 2022

- Architected a unified, modular ROS2 framework for a voice-controlled UR3e manipulator in industrial pick-and-place tasks for the ICRA 2022 Cocobots HRI project; the framework is currently leveraged by an Airbus subsidiary and academic researchers at the University of Potsdam and INRIA, who utilized the computer vision and NLP components, respectively.

Conbotics

Berlin, Germany

Robotics Software Engineer

Apr 2021 – Feb 2022

- As first employee, architected the complete MVP stack for the robot's mobile base - including autonomous navigation, computer vision, PID control and mission manager (FSM) - which successfully transitioned to market.

Airbond (Aerospace R&D)

Newport, Wales

Mechanical Design Trainee

Dec 2018 – Jun 2019

- Engineered aerospace composite splicer prototypes using SolidWorks and Industrial 3D Printing (HP Jet Fusion); work directly contributed to the company securing the Queen's Award for Enterprise in Innovation, with an anticipated annual value of £1M.

EDUCATION

KTH Royal Institute of Technology / TU Berlin

Stockholm / Berlin

MSc in Autonomous Systems (Grade: 1.7 / Top 5% GPA: 3.6/4.0)

2019 – 2022

- Specialized in applications of robotics and autonomous systems. Minor in innovation and entrepreneurship.
- EIT Digital Excellence Full Scholarship & Leventis Foundation Excellence Full Scholarship.

National Technical University of Athens (NTUA)

Athens, Greece

Diploma in Mechanical Engineering (Grade: 8.3/10 - Top 10% GPA: 3.5/4.0)

2013 – 2018

- 5 years program (300 ECTS) - Bachelor and MEng equivalent. Specialized in mechanical design, robotics and control systems.
- Thesis on Human–Autonomous Vehicle Interaction: Cyprus Ministry of Education 1st Prize NTUA 2nd Prize for Most Innovative Project.

LEADERSHIP & HONORS

- **UN Youth Representative:** Recognized as one of the 100 Young Climate Leaders globally; part of the youth OGCI taskforce.
- **Scholarships:** EIT Digital Excellence Full Scholarship and Leventis Foundation Excellence Full Scholarship.
- **Awards:** Cyprus Ministry of Education first prize award for best student project; NTUA's second prize award for the most innovative project.
- **Competitions:** Climate Launchpad Cyprus Finalist (2020)- reYOUse: Waste management web solution; Hult Prize Competition Greece Finalist (2017)
- EIT Digital Alumni and EIT Climate-KIC Alumni.

EXTRACURRICULAR TRAINING

- **Internships:** Research Assistant on Vehicle - Pedestrian Interaction (The Cyprus Institute, 2018); Research Assistant on Vehicle - Pedestrian Interaction Simulation (CYENS Research Centre, 2019); Sustainable Business Development (Pioneers into Practice Internship, 2020).
- **Technical Training:** Swarm Robotics Advanced Course (Delft University); Smart Mobility Summer School (EIT Digital - University of Trento - Developed a wagon tracking for Mercitalia); Climate Innovation Summer School (EIT Climate KIC - University of Copenhagen, University of Hamburg, ETH Zurich - Developed a carbon capture solution); AI Agents Course (HuggingFace, 2025); Complete Self-Driving Car Course (Udemy, 2021); Advanced 3D Printing Training (HP, 2019).
- **Military Service:** Telecommunication Officer Designate (National Guard of Cyprus, 2011 - 2013); achieved the highest non-commissioned rank based on IQ, academic merit, and physical testing; led a cohort of 40 soldiers; managed the communications department; authored telecom operations manual.

VOLUNTARY EXPERIENCE

- **Owner - Memorandum:** Maintained a page to promote youth opportunities. That was the successor of Youth Hero page that served the same purpose.
- **Main Coordinator - Green Deal 4 Cyprus:** Promoted social and environmental transition in Cyprus through the European Green Deal.
- **Project Manager - Fotspår:** Managed partners' collaboration in Sweden.
- **Main Organizer - BEST:** Organized advanced courses on sustainable transportation and member of the core team for European BEST Engineering Competition Greece.
- **Other:** participated in 3 Erasmus+ youth exchanges; co-authored the Ungdomsoen Manifesto; Co-authored the Sustainable Transportation Manifesto - Generation Climate Europe; Maintainer of a ride-sharing platform: Car Share Cyprus platform.

INTERESTS

Sports: Classic Marathon runner, hiking, climbing, tennis, and cycling.

Arts: Improv theatre, DJing, music production, guitar, and salsa/house dance.

Personal: Blood donor; Book lover.